

RIGIDITY AND UNIVERSAL CONTRACTIONS OF HARD-SPHERE KINETIC COTANGENTS

QIAN QI

ABSTRACT. We construct genuine positive microscopic phase labels from separated components of an already proved platform rate, retain an explicit boundary band, and use the same label at every observable projection. Restricted and conditional component laws, normalized component pressures, phase-local coefficients and exact positive mixture contraction yield the nonconvex physical rate without Riesz projections as probabilities or double counting phase costs.

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1. ROUND-THIRTEEN POSITIVE CLOSURE: CANONICAL POSITIVE BASINS AND PHASE-MIXTURE CONTRACTIONS

Phase labels are measurable events on the original microscopic sample space. They are not Riesz projections. A finite family of separated compact components of a platform rate is converted into disjoint positive basins, with one explicit boundary component that is retained at its actual exponential cost.

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1.1. Canonical basin labels. Let X_n take values in a Polish platform path space $\mathcal{X}$ and satisfy a good LDP with rate I . Let $M : \mathcal{X} \rightarrow \mathbb{R}^d$ be a continuous order parameter. On a regular finite-phase chart assume that the relevant compact sublevel contains finitely many disjoint compact components $K_1, \dots, K_J$ separated by distance $4\delta > 0$, and that every finite-rate point in the chart lies either in one basin or in the declared boundary band. These components are obtained from the A3 or B2 platform rate, not from an operator spectral projection.

Define the hard measurable label

$$J_n(x) = \begin{cases} j, & d(M(x), K_j) < \delta \text{ and } j \text{ is the unique nearest component,} \\ \partial, & \text{otherwise.} \end{cases}$$

Let $B_j = \{J_n = j\}$ and $B_\partial = \{J_n = \partial\}$; the sets do not depend on the observable projection level.

Theorem 1.1 (Exact positive microscopic disintegration). *For every n ,*

$$\mathbb{P}_n = \sum_{j=1}^J w_{n,j} \mathbb{P}_{n,j} + w_{n,\partial} \mathbb{P}_{n,\partial},$$

where $w_{n,j} = \mathbb{P}_n(B_j)$ and $\mathbb{P}_{n,j} = \mathbb{P}_n(\cdot \mid B_j)$. All weights are nonnegative, all component laws are probabilities, and the same label is available in every finite projection. The boundary component is never declared superexponential unless its rate actually is infinite.

Proof. The nearest-component rule is a Borel partition of the original sample space. The displayed identity is the ordinary law of total probability. Since J_n is defined from the full path order parameter, every projected label is its contraction and is automatically projectively compatible. No positivity property of a nonleading spectral projection is used. $\square$

1.2. Restricted and conditional LDPs. Put

$$\alpha_j = \inf_{x \in \bar{B}_j} I(x), \quad \alpha_\partial = \inf_{x \in \bar{B}_\partial} I(x).$$

Assume the basin boundaries are I -continuity sets and that every finite-rate point in B_j has a recovery sequence remaining in B_j .

Theorem 1.2 (Positive component LDP). *The unnormalized measures $\mathbb{P}_n(\cdot; B_j)$ satisfy the LDP with rate I restricted to $\bar{B}_j$. If $w_{n,j} > 0$ and $-n^{-1} \log w_{n,j} \rightarrow \alpha_j$, the conditional laws $\mathbb{P}_{n,j}$ satisfy a good LDP with*

$$I_j(x) = \begin{cases} I(x) - \alpha_j, & x \in \bar{B}_j, \\ +\infty, & \text{otherwise.} \end{cases}$$

The same theorem applies to the boundary label. Contracting the exact label gives

$$I(x) = \min_{j \in \{1, \dots, J, \partial\}} \{\alpha_j + I_j(x)\},$$

with no converification.

Proof. For a closed set F , apply the platform LDP upper bound to $F \cap \overline{B}_j$. For an open set and a point in B_j , use the assumed interior recovery sequence; boundary points follow by lower-semicontinuous closure. Taking the whole basin gives the weight exponent α_j . Dividing by the weight subtracts exactly that exponent and proves the conditional rate. The contraction principle for the finite positive label then gives the pointwise minimum. $\square$

1.3. Normalized component pressures. Define

$$\tilde{Q}_{n,j}(\theta) = \frac{1}{n} \log \mathbb{E}_n \left[e^{n\langle \theta, X_n \rangle} \mid B_j \right],$$

while the unnormalized restricted pressure is

$$Q_{n,j}^{\text{un}}(\theta) = \frac{1}{n} \log \mathbb{E}_n \left[e^{n\langle \theta, X_n \rangle}; B_j \right].$$

Proposition 1.3 (Single-count phase pressure). *Whenever the component Laplace principles hold,*

$$Q_j^{\text{un}}(\theta) = -\alpha_j + \tilde{Q}_j(\theta), \quad Q(\theta) = \max_{j \in \{1, \dots, J, \partial\}} \{-\alpha_j + \tilde{Q}_j(\theta)\}.$$

The phase cost is counted once.

Proof. The first identity is the exact factorization of the restricted expectation into its weight and conditional expectation. Sum the positive restricted expectations and take $n^{-1} \log$ to obtain the maximum. $\square$

1.4. Phase-local analytic charts. Inside a basin suppose the platform tilted rate has one unique regular minimizer separated from the basin boundary by a rate gap $g > 0$. Let χ_j be a smooth positive cutoff equal to one in a neighborhood of that minimizer and supported in B_j .

Theorem 1.4 (Localized zero-free and response chart). *On a sufficiently small complex source ball, the localized partition function with cutoff χ_j equals the unlocalized simple phase contribution plus an error $O(e^{-gn/2})$ relative to that contribution. Hence it is zero-free there for large n , has the same limiting analytic pressure, and its first two derivatives give the phase mean and covariance. The boundary label has its own real LDP and is not assigned a fictitious analytic chart.*

Proof. The platform source theorem gives a simple analytic phase contribution near the unique interior minimizer. On the support of $1 - \chi_j$, the real part of the tilted rate is larger by g ; compact complex perturbations reduce the gap by at most $g/2$. The relative error is therefore $O(e^{-gn/2})$ uniformly on the complex ball. Rouché's theorem preserves zero-freeness and the limiting pressure. Cauchy estimates give derivative convergence. $\square$

1.5. Shell, Gaussian, and dynamic mixtures. Assume the A2 or B1 local coefficient theorem holds uniformly on every active regular basin. Let $c_{n,j}(a)$ be the corresponding shell coefficient.

Theorem 1.5 (Exact phase-conditioned shell mixture). *Conditioning the original law on a regular shell gives phase weights proportional to*

$$w_{n,j}c_{n,j}(a)e^{-nI_j(a)}$$

after including the boundary label. Every subsequential limit is the corresponding mixture of the component conditioned laws. If one exponent is strictly minimal, that phase is selected; at a tie, subexponential weights and coefficients determine the limiting mixture.

Proof. Intersect the exact positive disintegration of Theorem 1.1 with the shell. Apply the uniform component coefficient theorem to every term and retain the boundary term at its actual rate. Divide by the sum. This is an exact finite positive mixture, so no signed cancellation or double-counted phase cost occurs. $\square$

Theorem 1.6 (Labelled commutation theorem). *On the joint labelled law, typed observable contraction, primitive-shell conditioning, finite-dimensional Gaussian scaling, and the component dynamic semigroups commute with the finite label disintegration. After the limits are proved componentwise, forgetting the label gives the positive mixture and the minimum rate of Theorem 1.2.*

Proof. All four operations are applied to the same positive measures $w_{n,j}\mathbb{P}_{n,j}$. Contraction is push-forward, conditioning is restriction and normalization, Gaussian scaling is another push-forward, and dynamic evolution acts linearly on the component laws before the nonlinear logarithm is taken. Finite summation commutes with these exact operations; uniform component limits permit passage to the limit. Forgetting the label is an ordinary contraction. $\square$

Theorem 1.7 (Round-thirteen D1 closure). *Every regular finite-phase chart of the Sinai or hard-sphere platform admits a canonical positive basin label on the original microscopic sample space, good component and boundary LDPs, normalized component pressures, phase-local response charts, and exact shell/Gaussian/dynamic mixture formulas. No nonleading Riesz projection is interpreted as a probability.*

Proof. Combine Theorems 1.1, 1.2, 1.4, 1.5, and 1.6, with Proposition 1.3. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-thirteen dependency ledger. Repository source, proof-structure, direct-counterexample, and

build gates are internal reproducibility checks and do not replace independent external mathematical review.

PEKING UNIVERSITY, BEIJING 100871, CHINA

Email address: `qiqian@pku.edu.cn`